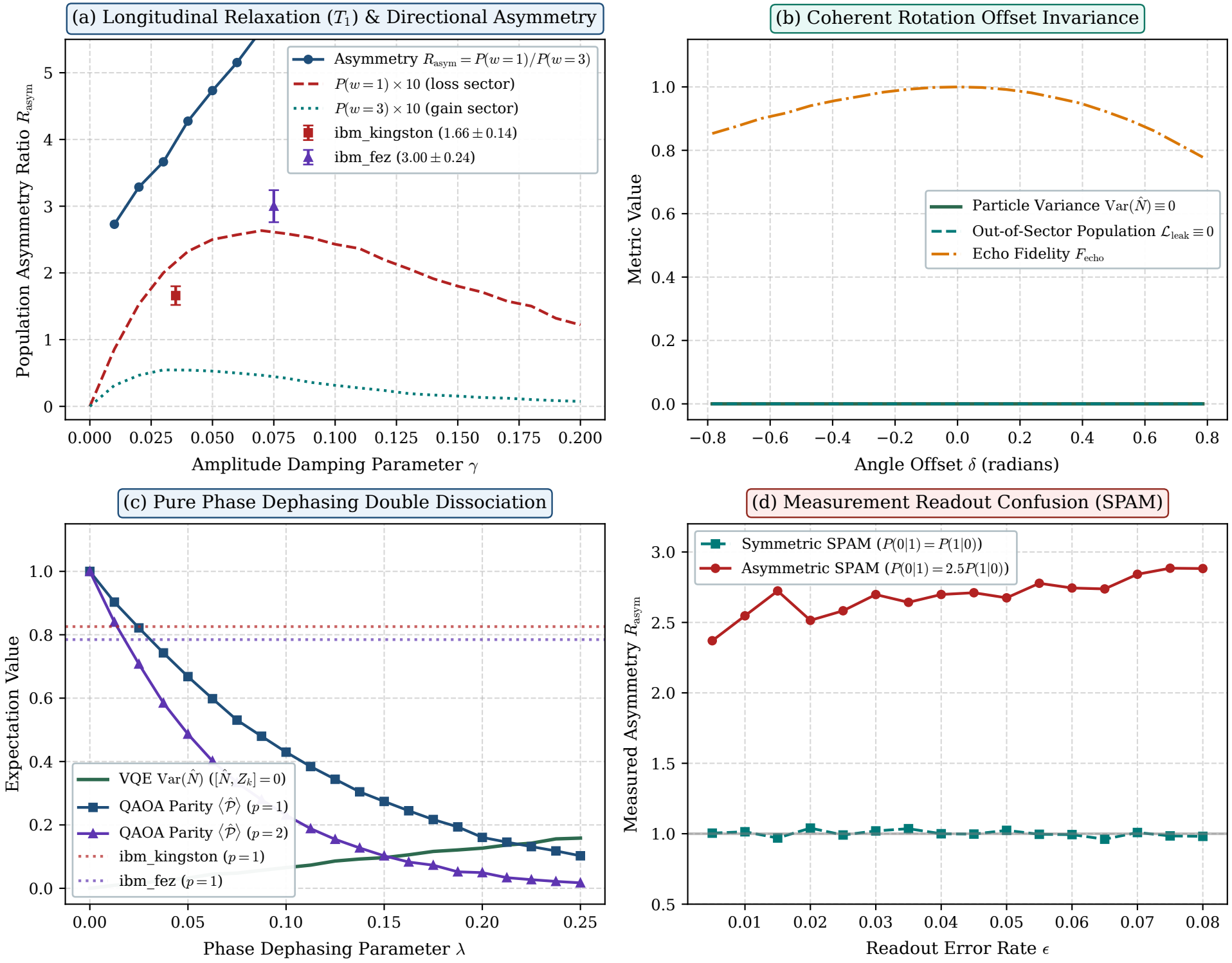
(a) Longitudinal Relaxation (T_1) & Directional Asymmetry
Asymmetry $R_{\text{asym}} = P(w=1)/P(w=3)$
$P(w=1) \times 10$ (loss sector)
$P(w=3) \times 10$ (gain sector)
ibm_kingston (1.66 ± 0.14)
ibm_fez (3.00 ± 0.24)
Population Asymmetry Ratio R_{asym}
Amplitude Damping Parameter γ
(b) Coherent Rotation Offset Invariance
Particle Variance $\text{Var}(\hat{N}) \equiv 0$
Out-of-Sector Population $\mathcal{L}_{\text{leak}} \equiv 0$
Echo Fidelity F_{echo}
Metric Value
Angle Offset δ (radians)
(c) Pure Phase Dephasing Double Dissociation
VQE $\text{Var}(\hat{N})$ ($[\hat{N}, Z_k] = 0$)
QAOA Parity $\langle \hat{P} \rangle$ ($p=1$)
QAOA Parity $\langle \hat{P} \rangle$ ($p=2$)
ibm_kingston ($p=1$)
ibm_fez ($p=1$)
Expectation Value
Phase Dephasing Parameter λ
(d) Measurement Readout Confusion (SPAM)
Symmetric SPAM ($P(0|1) = P(1|0)$)
Asymmetric SPAM ($P(0|1) = 2.5P(1|0)$)
Measured Asymmetry R_{asym}
Readout Error Rate ϵ